

Global El Niño Monitoring and Indian Monsoon Impact Assessment Using Google Earth Engine (1982–2026)

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Project Overview

The El Niño-Southern Oscillation (ENSO) is one of the most influential climate phenomena affecting global weather patterns. Variations in sea surface temperatures (SST) across the equatorial Pacific Ocean influence rainfall, drought occurrence, agriculture, and ecosystem productivity across many regions of the world, including India.

This project investigates the relationship between ENSO conditions and Indian monsoon variability using satellite observations and climate datasets processed entirely within Google Earth Engine (GEE). Historical ENSO events were analyzed from 1982 to 2025, followed by rainfall and vegetation impact assessments across India. A monitoring dashboard was also developed to track current ENSO and monsoon conditions during 2026.

Objectives

The primary objectives of this project were:

1. Monitor ENSO conditions using the Niño 3.4 Index.
 2. Build a historical ENSO Event Catalog from 1982–2025.
 3. Identify and compare major El Niño events.
 4. Assess Indian monsoon rainfall response during El Niño years.
 5. Evaluate vegetation response using NDVI.
 6. Develop a near-real-time monsoon monitoring dashboard for 2026.
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Study Area

ENSO Monitoring Region

Niño 3.4 Region

Latitude:

5°N to 5°S

Longitude:

170°W to 120°W

This region is widely used by climate scientists to monitor El Niño and La Niña conditions through sea surface temperature anomaly analysis.

Impact Assessment Region

Country:

India

Boundary Source:

FAO GAUL Administrative Boundaries

Datasets Used

NOAA Optimum Interpolation Sea Surface Temperature (OISST) Version 2.1

Dataset:

NOAA/CDR/OISST/V2_1

Purpose:

- Sea Surface Temperature Monitoring
- SST Anomaly Analysis
- Niño 3.4 Index Calculation
- ENSO Classification

Spatial Resolution:

0.25°

Temporal Resolution:

Daily

CHIRPS Daily Precipitation

Dataset:

UCSB-CHG/CHIRPS/DAILY

Purpose:

- Indian Monsoon Rainfall Analysis
- El Niño Rainfall Impact Assessment

Spatial Resolution:

0.05°

Temporal Resolution:

Daily

MODIS Vegetation Indices

Dataset:

MODIS/061/MOD13Q1

Purpose:

- Vegetation Health Monitoring
- NDVI Analysis
- Ecosystem Response Assessment

Spatial Resolution:

250 m

Temporal Resolution:

16-Day Composite

FAO GAUL Administrative Boundaries

Purpose:

- India Boundary Extraction
 - Spatial Clipping
 - Area-Based Statistics
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Methodology

Phase 1: ENSO Monitoring

A Niño 3.4 monitoring region was created within the equatorial Pacific Ocean.

Monthly SST anomalies were extracted from NOAA OISST data.

Monthly Niño 3.4 values were calculated by averaging SST anomalies within the Niño 3.4 region.

Phase 2: ENSO Event Catalog Development

Monthly Niño 3.4 values were generated for each year from 1982–2025.

For every year the following indicators were calculated:

- Mean Niño Index
- Peak Niño Index
- Peak Month
- Strong Event Months ($>1.5^{\circ}\text{C}$)
- Extreme Event Months ($>2.0^{\circ}\text{C}$)
- Onset Rate

These indicators were compiled into a custom ENSO Event Catalog.

Phase 3: Historical Event Comparison

Major ENSO years were analyzed:

- 1982
- 1997
- 2015
- 2023

Monthly Niño 3.4 evolution was compared to identify differences in onset timing, peak intensity, and duration.

Phase 4: Indian Monsoon Rainfall Assessment

Monsoon season was defined as:

June–September (JJAS)

CHIRPS rainfall data were aggregated to produce seasonal rainfall totals.

Two composite datasets were generated:

Strong El Niño Years:

- 1982
- 1997
- 2015
- 2023

Neutral Years:

- 1984
- 1985
- 1990
- 2001
- 2013

Rainfall difference maps were generated by subtracting neutral-year rainfall from strong El Niño rainfall.

Phase 5: Vegetation Response Assessment

MODIS NDVI data were used to evaluate vegetation conditions.

Strong El Niño Years:

- 2015
- 2023

Neutral Years:

- 2001
- 2013
- 2014
- 2017
- 2018

Average NDVI composites were generated and compared to identify vegetation responses to ENSO conditions.

Phase 6: 2026 Monsoon Monitoring Dashboard

A monitoring dashboard was developed using:

- Current SST Anomalies
- Niño 3.4 Index
- Cloud Cover Monitoring
- Rainfall Monitoring

- Historical ENSO Comparisons

The dashboard provides a near-real-time view of current climate conditions.

Results

Historical ENSO Analysis

Years Analyzed:

44

Major El Niño Events Identified:

- 1982
- 1997
- 2015
- 2023

Peak Niño Values:

1982:
2.32°C

1997:
2.35°C

2015:
2.97°C

2015 was identified as the strongest El Niño event within the study period.

Rainfall Impact Assessment

Average Monsoon Rainfall Change:

-95 mm

Average Monsoon Reduction:

-10.5%

The strongest rainfall deficits were observed across central and peninsular India.

Vegetation Response Assessment

Average NDVI Change:

+0.039

Results indicate that vegetation response does not directly mirror rainfall reductions at the national scale.

Potential factors include:

- Irrigation
 - Groundwater availability
 - Agricultural practices
 - Regional climate variability
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Current 2026 Conditions

Current Niño 3.4 Index:

+1.65°C

Classification:

Strong El Niño

Current Rainfall Monitoring:

7-Day Average:

18.86 mm

30-Day Average:

41.12 mm

Cloud cover observations indicate active monsoon development across several regions of India.

Deliverables

Generated Outputs:

1. ENSO Event Catalog (CSV)
2. Historical ENSO Comparison Charts
3. SST Anomaly Maps

4. Rainfall Composite Maps
 5. Rainfall Impact Maps
 6. NDVI Composite Maps
 7. NDVI Impact Maps
 8. 2026 Monsoon Monitoring Dashboard
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Tools and Technologies

- Google Earth Engine
 - Remote Sensing
 - Climate Analytics
 - Time Series Analysis
 - Raster Processing
 - Spatial Statistics
 - Environmental GIS
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Conclusion

This project demonstrates the application of Google Earth Engine for large-scale climate analysis and environmental monitoring.

The analysis revealed that strong El Niño events are associated with an average reduction of approximately 95 mm (10.5%) in Indian monsoon rainfall. Vegetation response exhibited greater complexity, highlighting the influence of irrigation and land management practices.

The developed workflow integrates satellite observations, climate datasets, and geospatial analysis to create a scalable framework for monitoring ENSO conditions and assessing their impacts on rainfall and vegetation across India.